

OpenStax Calculus Volume 2

Section 2.5 Exercises: Physical Applications

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For the following exercises, find the work done.

1. Find the work done when a constant force $F = 12$ lb moves a chair from $x = 0.9$ to $x = 1.1$ ft.
2. How much work is done when a person lifts a 50 lb box of comics onto a truck that is 3 ft off the ground?
3. What is the work done lifting a 20 kg child from the floor to a height of 2 m? (Note that a mass of 1 kg weighs 9.8 N near the surface of the Earth.)
4. Find the work done when you push a box along the floor 2 m, when you apply a constant force of $F = 100$ N.
5. Compute the work done for a force $F = 12/x^2$ N from $x = 1$ to $x = 2$ m.
6. What is the work done moving a particle from $x = 0$ to $x = 1$ m if the force acting on it is $F = 3x^2$ N?

For the following exercises, find the mass of the one-dimensional object.

7. A wire that is 2 ft long (starting at $x = 0$) and has a density function of $\rho(x) = x^2 + 2x$ lb/ft
8. A car antenna that is 3 ft long (starting at $x = 0$) and has a density function of $\rho(x) = 3x + 2$ lb/ft
9. A metal rod that is 8 in. long (starting at $x = 0$) and has a density function of $\rho(x) = e^{(1/2)x}$ lb/in.
10. A pencil that is 4 in. long (starting at $x = 2$) and has a density function of $\rho(x) = 5/x$ oz/in.
11. A ruler that is 12 in. long (starting at $x = 5$) and has a density function of $\rho(x) = \ln(x) + (1/2)x^2$ oz/in.

For the following exercises, find the mass of the two-dimensional object that is centered at the origin.

12. An oversized hockey puck of radius 2 in. with density function $\rho(x) = x^3 - 2x + 5$
13. A frisbee of radius 6 in. with density function $\rho(x) = e^{-x}$
14. A plate of radius 10 in. with density function $\rho(x) = 1 + \cos(\pi x)$

15. A jar lid of radius 3 in. with density function $\rho(x) = \ln(x + 1)$
16. A disk of radius 5 cm with density function $\rho(x) = \sqrt{3x}$
17. A 12-in. spring is stretched to 15 in. by a force of 75 lb. What is the spring constant?
18. A spring has a natural length of 10 cm. It takes 2 J to stretch the spring to 15 cm. How much work would it take to stretch the spring from 15 cm to 20 cm?
19. A 1-m spring requires 10 J to stretch the spring to 1.1 m. How much work would it take to stretch the spring from 1 m to 1.2 m?
20. A spring requires 5 J to stretch the spring from 8 cm to 12 cm, and an additional 4 J to stretch the spring from 12 cm to 14 cm. What is the natural length of the spring?
21. A shock absorber is compressed 1 in. by a weight of 1 t. What is the spring constant?
22. A force of $F = 20x - x^3$ N stretches a nonlinear spring by x meters. What work is required to stretch the spring from $x = 0$ to $x = 2$ m?
23. Find the work done by winding up a hanging cable of length 100 ft and weight-density 5 lb/ft.
24. For the cable in the preceding exercise, how much work is done to lift the cable 50 ft?
25. For the cable in the preceding exercise, how much additional work is done by hanging a 200 lb weight at the end of the cable?
26. [T] A pyramid of height 500 ft has a square base 800 ft by 800 ft. Find the area A at height h . If the rock used to build the pyramid weighs approximately $w = 100 \text{ lb/ft}^3$, how much work did it take to lift all the rock?
27. [T] For the pyramid in the preceding exercise, assume there were 1000 workers each working 10 hours a day, 5 days a week, 50 weeks a year. If the workers, on average, lifted 10 100 lb rocks 2 ft/hr, how long did it take to build the pyramid?
28. [T] The force of gravity on a mass m is $F = -(GMm)/x^2$ newtons. For a rocket of mass $m = 1000$ kg, compute the work to lift the rocket from $x = 6400$ to $x = 6500$ km. State your answers with three significant figures. (Note: $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$ and $M = 6 \times 10^{24} \text{ kg}$.)
29. [T] For the rocket in the preceding exercise, find the work to lift the rocket from $x = 6400$ to $x = \infty$.
30. [T] A rectangular dam is 40 ft high and 60 ft wide. Assume the weight density of water is 62.5 lbs/ft^3 . Compute the total force F on the dam when
 - (a) the surface of the water is at the top of the dam and
 - (b) the surface of the water is halfway down the dam.
31. [T] Find the work required to pump all the water out of a cylinder that has a circular base of radius 5 ft and height 200 ft. Use the fact that the density of water is 62 lb/ft^3 .
32. [T] Find the work required to pump all the water out of the cylinder in the preceding exercise if the cylinder is only half full.
33. [T] How much work is required to pump out a swimming pool if the area of the base is 800 ft^2 , the water is 4 ft deep, and the top is 1 ft above the water level? Assume that the density of water is 62 lb/ft^3 .

34. A cylinder of depth H and cross-sectional area A stands full of water at density ρ . Compute the work to pump all the water to the top.
35. For the cylinder in the preceding exercise, compute the work to pump all the water to the top if the cylinder is only half full.
36. A cone-shaped tank has a cross-sectional area that increases with its depth: $A = (\pi r^2 h^2)/H^2$. Show that the work to empty it is half the work for a cylinder with the same height and base.